

deepiri-fuselk

Fusion Unified Simulation, ELM Learning & Kinetics

Team Deepiri

Version 0.2.0

Generated from deepiri-fuselk

Keywords fusion, tokamak, elm, plasma, simulation, reinforcement-learning

Deepiri Research

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Abstract

Fusion Unified Simulation, ELM Learning & Kinetics — open tokamak plasma control platform **Fusion Unified Simulation, ELM Learning & Kinetics**

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Introduction

[!Python](https://img.shields.io/badge/python-3.11%2B-blue?logo=python)(https://python.org)
[!License: Apache 2.0](https://img.shields.io/badge/License-Apache%202.0-blue)(LICENSE) Key
topics: fusion, tokamak, elm, plasma, simulation, reinforcement-learning. Source:
<https://github.com/Team-Deepiri/deepiri-fuselk>.

****Fusion Unified Simulation, ELM Learning & Kinetics****

System Architecture

deepiri-fuselk Architecture

Modules

Module	Purpose
`physics/`	Oil-water coupled PDE system, energy balance
`barrier/`	Vapor dynamics, tritium breeding blanket
`helix/`	Boozer mapping, phase-locked tracker, HQRM
`focal/`	Focal heat maps, spiral attention
`muon/`	Rate network, photon/proton stripping
`control/`	Traffic router, vent RL env, watchdog
`sim/`	Digital twin, synthetic data, domain randomizer
`viz/`	Dash dashboard, HELIX/traffic viewers

Data Flow

Diagnostics → HELIX/HQRM → Focal Heat Map → RL Control → Actuators
↓
Physics PDE ← Breeding Blanket ← Muon Cycle

See docs/theory/ for governing equations.

Package	Role
deepiri_fuselk	Deepiri Fuselk
deepiri_fuselk.barrier	Barrier
deepiri_fuselk.control	Control
deepiri_fuselk.data	Data
deepiri_fuselk.focal	Focal
deepiri_fuselk.helix	Helix

Package	Role
deepiri_fuselk.models	Models
deepiri_fuselk.muon	Muon
deepiri_fuselk.physics	Physics
deepiri_fuselk.sim	Sim
deepiri_fuselk.viz	Viz
src.deepiri_fuselk	Deepiri Fuselk
src.deepiri_fuselk.barrier	Barrier
src.deepiri_fuselk.control	Control
src.deepiri_fuselk.data	Data
src.deepiri_fuselk.focal	Focal
src.deepiri_fuselk.helix	Helix
src.deepiri_fuselk.models	Models
src.deepiri_fuselk.muon	Muon
src.deepiri_fuselk.physics	Physics
src.deepiri_fuselk.sim	Sim
src.deepiri_fuselk.viz	Viz

Theoretical Foundation

Helix Hqrm

HELIX / HQRM

HELIX Engine

Helical Extraction & Locked-In Isotropy eXclusion:

- Phase-locked sampling at O-point rotation
- Boozer coordinate unwrap
- Spiral attention along field lines
- Focal singularity heat map output

HQRM

Helical Quadtree Rotational Mapper:

- Recursive side-by-side square subdivision
- Local rotation aligned to field line pitch
- 7x7 kernel lock when variance < 7%

Muon Rate Network

Muon Catalytic Rate Network

Populations

$$\delta = \left(\frac{\partial \phi}{\partial \lambda} \right)^{24}$$

Extraction criterion

$$|\lambda| > \frac{\delta}{\epsilon} + \text{Re}(\lambda)$$

Experiments

ID	Category	Status	Description
elm_helix_denoising	diagnostics	simulation	Phase-locked O-point tracking with HQRM 7x7 kernel lock
oil_water_barrier	materials	simulation	Coupled plasma/vapor PDE with tritium breeding
tritium_vapor_extraction	fuel_cycle	simulation	Peclet-driven outward tritium sweep in lithium vapor blanket
muon_photon_stripping		speculative	X-ray photoelectric + vortex OAM muon reactivation
brine_wall_coating	materials	speculative	Simulation-only study of salinity barrier coatings (not validated)
vent_circularization_rl	control	simulation	Hierarchical RL for divertor heat flux symmetrization
plasma_traffic_routing	control	simulation	Real-time divertor heat flux routing and congestion detection

Recent Development

- **Jupyter notebook suite** — six vision/math/experiment notebooks under `notebooks/` (HELIX, oil–water PDEs, muon cycle, Venturi control, ReactorCell digital twin)
- Jupyter dev dependencies (`jupyter`, `ipykernel`, `matplotlib`, `nbformat`)
- **FusionKPIs** composite progress score for reproducible benchmarking
- **Trainable ELMPredictor** — logistic regression on labeled synthetic shot corpus with save/load
- **DisruptionDetector** — fuses HELIX, ELM, and MHD stability into actuation recommendations
- **Shot corpus generator** for supervised ELM training

References

- [1] deepiri-fuselk source repository. <https://github.com/Team-Deepiri/deepiri-fuselk>
- [2] pdf-genesis research paper synthesizer.